

Does a Physiology Engine Improve Alerts in Heart Failure Home Monitoring? A Digital Twin Case Study

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Abstract—Home monitoring gives heart-failure (HF) patients daily measurements but no physiological interpretation. We built a digital twin to test whether a mechanistic physiology engine can supply one and improve alerts. A Random Forest reads a 21-day window of six daily trends and routine clinical values, estimates a deterioration scenario and its severity, and drives the Kitware Pulse engine, whose simulated hemodynamics are scored by an interpretable risk score; engine state is carried from day to day. On a pre-specified cohort of ten scripted 21-day patient stories (six seeds), the classifier’s day-by-day scenario label was correct on only 17.6% of days, collapsing to “deconditioning” on 72%. The deployed Pulse-based alert detected 70% of deteriorating series with 8.7 false alerts per 100 patient-days. However, the classifier’s own severity, with a threshold selected on development seeds, detected all held-out deteriorating series at the same held-out false-alert rate (11.6 per 100 patient-days) as the Pulse system, which detected 10 of 15. The full pipeline ran on real home data from 13 of 16 HF patients, but through an age-proxy body with default ejection fraction, and its scores did not predict real hospital outcomes beyond chance. In this case study the physiology engine therefore added interpretable hemodynamics but no measurable alerting benefit over a tuned classifier; we identify the label errors and scenario mis-specification that would need to be fixed for it to do so.

Index Terms—Digital twin, heart failure, home monitoring, wearable sensors, physiology simulation, Pulse engine, random forest, remote patient monitoring, early warning, reproducibility.

I. INTRODUCTION

Heart failure affects tens of millions of people worldwide [1], [2] and is defined by progression: patients decline over days to weeks rather than instantaneously. Current guidelines emphasize early recognition of congestion and decompensation [3], yet three structural gaps remain in practice. First, monitoring is episodic: patients are typically seen every few weeks, and a patient can move from stable to crisis between visits. Second, consumer wearables report numbers, not phys-

iology: a rising resting heart rate is measured, but whether it reflects benign variation or an early compensatory response to falling cardiac output is left to the patient. Third, threshold alarms (e.g., $\text{SpO}_2 < 92\%$) fire only after a value is already abnormal; they describe where a patient is, not where they are heading.

Remote monitoring has shown benefit when it captures physiology rather than symptoms alone, from structured tele-monitoring [4] to implanted pulmonary-artery pressure sensors [5], and multi-sensor wearables combined with machine learning have predicted HF hospitalization days in advance [6], [7]. These data-driven predictors, however, offer little mechanistic explanation of *why* an alert fired. Cardiac digital twins promise exactly that mechanistic layer [8]–[10], but most are built from imaging or invasive measurements fitted to one patient at a time [11], [12], and few are evaluated over weeks of everyday data.

This paper reports a case study of a twin that sits between these two traditions: it is driven by daily home-monitoring trends (wearable and home-device signals), interprets them through a mechanistic whole-body physiology engine (Kitware Pulse [14]), and produces interpretable risk, staging and forward projections. We use “twin” in a limited sense: the engine body is parameterized by population statistics and scaled by a patient’s ejection fraction and estimated severity, not fitted to the individual. Our contributions are:

- 1) A complete, open pipeline (wearable trends $\rightarrow$ ML scenario and severity estimation $\rightarrow$ Pulse simulation $\rightarrow$ interpretable risk scoring, staging and projection), deployed as a stateful service with day-to-day continuity of engine state (Sections III–V).
- 2) Engine-level findings for using Pulse as an HF twin: mapping ejection fraction onto the engine, a root-caused instability boundary when disease modifiers and exertion

TABLE I
POSITIONING AGAINST RELATED WORK

Work	Input data	Model	Personalization	Gap addressed here
Corral-Acero 2020 [8], Coorey 2022 [9] Gu 2025 [11]	Multimodal (vision/review) Clinical, echo, invasive	Mechanistic + statistical Lumped cardiovascular model + unsupervised ML	Conceptual Fitted per patient (343 HF)	No deployed system or longitudinal test Requires hospital measurements; no home monitoring
Geddes 2025 [12]	Imaging, invasive validation	3D CFD	Fitted per patient	Right HF only; not wearable-driven
Balasubramanyam 2025 [13] Stehlik 2020 (LINK-HF) [6]	ECG datasets, echo Wearable patch	Compliance model, Fourier ECG, 3D mesh ML (similarity-based)	Signal-level Patient baseline	Visualization and ECG fidelity, not deterioration tracking No mechanistic explanation of alerts
This work	Wearable trends + routine clinical	RF + Pulse engine + interpretable score	Scenario/severity-driven, EF-scaled	Mechanistic, wearable-driven, evaluated over 21 days

combine, and an exact state save/resume method (Section VI).

- 3) A layered evaluation centered on an ablation that asks whether the physiology layer improves alerting over the classifier alone (including a threshold tuned on development seeds and a label-only rule) and over a clinical weight rule, on a pre-specified cohort with held-out seeds, plus real-patient replay and real-outcome tests (Section VII). The answer, in this case study, is no.

II. RELATED WORK

Cardiac digital twins. Corral-Acero *et al.* [8] framed the digital twin as the route to precision cardiology, Coorey *et al.* [9] reviewed the emerging health-twin field for cardiovascular disease, and Niederer *et al.* [10] argued that twins must move from artisanal, single-patient builds to scalable, industrial pipelines. Recent HF-specific twins fit mechanistic cardiovascular models to measured data: Gu *et al.* [11] fitted twins to 343 HF patients and used the fitted parameters for interpretable phenotyping and prognosis, and Geddes *et al.* [12] built CFD-based twins that non-invasively estimate markers of right HF. Balasubramanyam *et al.* [13] proposed a cardiovascular twin combining time-varying compliance, Fourier-based ECG reconstruction and 3D anatomical modeling.

Wearable HF prediction. The LINK-HF study [6] combined a wearable patch with machine learning to predict HF hospitalization with a median lead time of several days, and Gautam *et al.* [7] surveyed AI, wearables and remote monitoring in HF. Invasive hemodynamic monitoring reduced HF hospitalization in the CHAMPION trial [5].

Risk scores. The MAGGIC score [17] is a widely used, externally validated HF survival score, and the LACE index [18] is a general readmission score; both serve as external benchmarks here.

Positioning. Table I contrasts these approaches. Mechanistic twins personalize deeply but require hospital measurements; wearable predictors scale but are opaque. Our twin trades per-patient parameter fitting for deployability from home data, and adds a mechanistic layer that wearable-only predictors lack.

TABLE II
TECHNOLOGY STACK

Layer	Component
Physiology engine	Kitware Pulse 4.3.1 (kitware/pulse:4.3.1 image, amd64) [14]
Machine learning	scikit-learn 1.9.0 Random Forest [19], [20]; XGBoost 3.2.0 (secondary) [21]
Back end	Python 3.11, FastAPI, SQLAlchemy, Pydantic; SQLite (Postgres-portable)
Front end	React 19 + Vite; hand-built SVG charts
Deployment	Docker Compose (Postgres, back end, front end)
Testing	pytest (281 automated tests), Playwright browser checks, GitHub Actions CI

We claim novelty for the integrated pipeline and its evaluation, not for any single component.

III. SYSTEM OVERVIEW

Fig. 1 shows the pipeline. A patient record holds demographics and a clinical report (ejection fraction, NT-proBNP); daily wearable readings accumulate until a 21-day window is filled, which triggers one background assessment: (1) the scenario classifier and severity regressor (the “scenario model”); (2) construction of a Pulse patient and scenario file; (3) simulation inside the Pulse container with crash and timeout detection; (4) hemodynamic feature extraction; (5) risk scoring, NYHA staging and trend-based deterioration rate; and (6) three further Pulse runs at projected severities for 7, 14 and 30 days. All results are persisted, so every read endpoint is a database read and never blocks on the engine. Table II lists the technology stack.

IV. DATA

A. Reference data and synthetic training population

No real patient data were used for training. Instead, every distribution parameter used by the synthetic generator is traced to an aggregate statistic from a real source (Table III): MIMIC-IV HF admissions [22] for age, sex and laboratory values; the Chicco–Jurman HF clinical records [24] as the only source with ejection fraction; and NHANES for height, weight and BMI by sex. A generator produced $n = 2,000$ synthetic

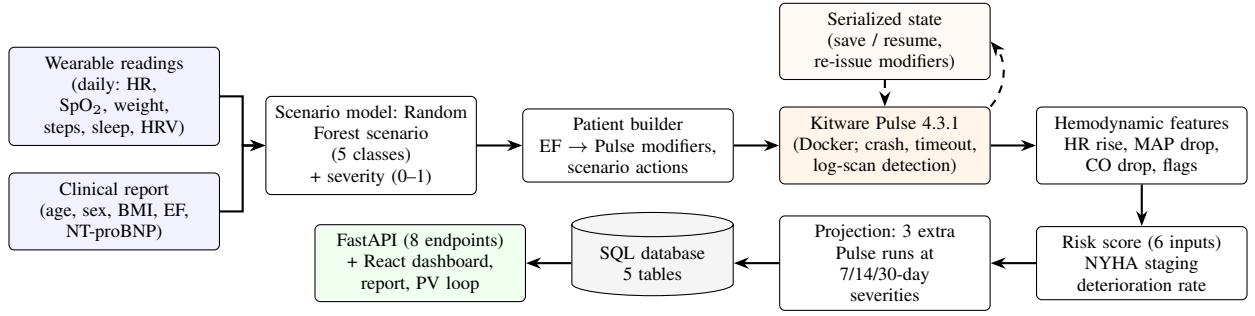


Fig. 1. System architecture. A filled 21-day wearable window triggers one background assessment; the dashed loop is the continuous-state mode, in which each day’s simulation resumes from the previous day’s serialized engine state (Section VI).

patients across five scenarios. Each patient receives a scenario and a severity $\sigma \sim U(0, 1)$ (capped at 0.15 for stable patients). Ejection fraction is drawn from the scenario’s EF profile and lowered by 12σ points; NT-proBNP is driven by a mix of EF deficit and severity scaled to age-specific diagnostic cut-offs, which makes BNP fall with EF and rise with severity. Each wearable signal v on day t then follows

$$v(t) = b_v + \Delta_{v,c} \sigma g_m(t/20) + \epsilon_{v,t}, \quad (1)$$

where b_v is a patient baseline (drawn from population statistics, or the patient’s own weight), $\Delta_{v,c}$ is the maximum change for scenario c (e.g., fluid overload: +4.5 kg weight, +12 bpm resting HR, −2.5 points SpO_2), g_m is a progression curve for mode m (deteriorating: linear, quadratic for acute deterioration; recovering: $\max(0, 1 - 1.15u)$; stable: 0), and $\epsilon_{v,t}$ is independent Gaussian noise with SD equal to 15% of the population SD (0.4 kg for weight). Signals are therefore correlated only through the shared σ and g_m , not through any physiological coupling. The deteriorating mode is chosen with probability $0.4 + 0.5\sigma$. This construction matters for interpreting results: a classifier trained on it learns to invert these deltas, so held-out accuracy measures recovery of the generator’s own labels, not clinical validity. Only aggregate statistics from the real sources are stored in the repository; raw files remain outside version control.

Two candidate sources were rejected after inspection: a 5,000-row “HF clinical records” variant turned out to be a resampled copy of the same ~ 300 patients (only 203 unique platelet values, 3,680 exact duplicate rows), and a public wearable dataset described as HF data contained general fitness data from 16 participants.

B. Scenario taxonomy

Five deterioration scenarios were fixed before modeling (Table IV). Each maps to a specific set of Pulse actions, chosen so that the simulated physiology differs between scenarios rather than merely their labels.

V. METHODS

A. Scenario model: classifier and severity regressor

Two Random Forests [19] (300 trees, default depth) share one 29-column feature matrix: five clinical features (age, sex,

TABLE III
DATASETS AND THEIR ROLES

Source	Role	Access
MIMIC-IV [22]	Reference statistics; real-outcome test (17,129 HF ICU admissions)	PhysioNet credentialed [23]
Chicco & Jurman [24]	EF, creatinine, sodium statistics	Public, CC BY 4.0
NHANES	Height/weight/BMI by sex	Public survey
PerHeart [25]	Real home data, 27 HF patients aged 67–94; pipeline replay	Ethics-approved, Zenodo
Zigong HF cohort [26]	Real-outcome tests (2,001 admissions)	PhysioNet, DUA
Multi-pathology BCG [27]	Measured SV/CO for 2 patients	figshare

TABLE IV
SCENARIO TAXONOMY AND PULSE REPRESENTATION

Scenario	Wearable signature	Pulse actions
stable	Flat trends	EF-derived modifiers only
fluid_overload	Rising weight, rising HR	Venous compliance reduced with severity
cardiac_stress	HR, HRV changes under exertion; preserved EF	Modifiers + Exercise
deconditioning	Falling steps, mild HR drift	Resistance/compliance modifiers; no exertion
acute_deterioration	Rapid multi-signal decline; reduced EF	Modifiers + HR multiplier + Exercise

BMI, EF, NT-proBNP) and, for each of six wearable signals, the first-7-day mean, last-7-day mean, their difference and the 21-day linear slope (24 features). The trend-mode column used during synthesis is dropped to prevent label leakage. Patients are split 70/15/15 (stratified by scenario) into training, validation and test sets.

B. Physiology engine and personalization

Pulse models the circulation as a lumped-parameter network, the hydraulic analogue of an electric circuit: nodes carry pressures, paths carry flows, vessel segments are resistors and elastic segments are capacitors, so that for each segment

$$Q = \frac{\Delta P}{R}, \quad C = \frac{dV}{dP}, \quad (2)$$

with arterial segments represented as Windkessel elements (a compliance in parallel with a resistance) [16]. Each heart chamber is a time-varying compliance whose inverse, the elastance, follows a model adapted from Stergiopoulos *et al.* [15]:

$$E_v(t) = (E_{\max,v} - E_{\min,v}) \frac{f(t)}{f_{\max}} + E_{\min,v}, \quad (3)$$

$$f(t) = \frac{(t/\alpha_1 T)^{n_1}}{1 + (t/\alpha_1 T)^{n_1}} \cdot \frac{1}{1 + (t/\alpha_2 T)^{n_2}}, \quad (4)$$

where $E_{\max,v}$ and $E_{\min,v}$ are the maximum and minimum ventricular elastances, T is the cardiac-cycle duration and $\alpha_1, \alpha_2, n_1, n_2$ are shape parameters; chamber compliance is $1/E_v(t)$. The cyclic change in compliance produces the pressure and volume changes that drive flow through the network, and stroke volume, cardiac output and arterial pressures are computed from the circuit quantities at each time step. Ejection fraction is therefore an *output* of this model, not an input.

We therefore map EF and severity onto Pulse’s CardiovascularMechanicsModification multipliers (stroke volume, systemic resistance and compliance, venous compliance, heart rate) and, for reduced EF, the binary ChronicVentricularSystolicDysfunction condition (a fixed elastance reduction). When the condition is applied, the continuous multiplier represents only the *additional* acute severity, because stacking both produced cardiovascular collapse in validation. Pulse rejects patients outside age 18–65 and BMI 16–30; demographics are clamped only at the engine boundary, so the patient’s true EF and severity still drive the simulation while the simulated body is a proxy.

C. Hemodynamic features and risk score

From each run we extract HR rise, MAP drop, cardiac-output (CO) drop, a compensation flag (stroke volume held $\geq 95\%$) and an instability flag ($\text{MAP}_{\text{end}} < 65$ mmHg). The primary risk score (Table V) normalizes each to $[0, 1]$ against a clinically anchored full scale and forms a weighted sum. During validation the score was structurally blind to fluid overload, whose danger appears as an already-congested baseline rather than an acute change, so a sixth input was added:

$$r = \max\left(\sum_i w_i s_i, s_{\text{baseline}}(\text{MAP}_{\text{start}})\right), \quad (5)$$

where s_i are the five acute sub-scores and s_{baseline} scores how far the simulated starting MAP lies below a healthy 92.5 mmHg anchor. Buckets are LOW ($r < 0.35$), MODERATE ($r < 0.65$) and HIGH; these cut-points are an engineering choice, not a clinical citation. We later found that the fluid-overload scenario itself is mis-specified (Section IX): it lowers venous compliance without adding circulating volume. The baseline-deficit term therefore compensates for a flawed scenario rather than reflecting correctly simulated congestion, and its improvement on the batch (30/30 LOW to 29/30 MODERATE) should be read in that light. An XGBoost regressor [21] on the same features is kept as a secondary comparison only

TABLE V
RISK-SCORE COMPONENTS AND CLINICAL ANCHORS

Component	Weight	Anchor (citation)
HR rise	0.15	NEWS2 heart-rate bands [28]
MAP drop	0.20	Healthy baseline to MAP 65 mmHg [29]
CO drop	0.20	30% decline = full scale [30], [31]
Compensation failure	0.15	Frank–Starling failure (binary)
Instability flag	0.30	MAP < 65 mmHg [29]
Baseline deficit	max()	Starting MAP below 92.5 mmHg anchor

TABLE VI
WHEN EACH SCORING ELEMENT WAS FIXED

Element	Status	Independent evaluation
Risk weights and anchors	Pre-specified (literature)	All later data
Bucket cut-points 0.35/0.65	Pre-specified (engineering)	All later data; not clinically validated
Baseline-deficit term	Post-hoc on 117-run batch	Scenario test, PerHeart run 3, MIMIC-IV, Zigong
NYHA feature removed	Post-hoc on live batch	45 new live patients; PerHeart
C3 component guard	Post-hoc on dev seeds	Held-out seeds 45–47
WATCH tier (decide_alert)	Post-hoc	Replay on both seed sets only
Scenario expectations	Pre-specified (git)	—

(5-fold CV MAE 0.089 ± 0.020 , R^2 0.828 ± 0.091 ; $n = 117$ is too small to make it primary).

D. Pre-specified and post-hoc design decisions

Several scoring elements were changed after seeing results. Table VI records when each was fixed relative to the data it was evaluated on, so that results on data used for tuning are not mistaken for out-of-sample evidence.

E. Staging, deterioration rate and projection

NYHA staging is rule-based: AHA/ACC structural criteria ($\text{LVEF} \leq 40\%$ or age-adjusted NT-proBNP above cut-off) [3] gate whether symptoms have a structural basis, and the simulated exertion response places the patient in classes I–IV. A deterioration rate is computed from per-vital slopes normalized to population standard deviations per day; projection re-runs the full Pulse pipeline at linearly extrapolated severities for 7, 14 and 30 days.

F. Service, alerts and continuous state

The FastAPI service exposes eight endpoints over five tables (patients, clinical reports, wearable readings, simulation runs, risk assessments). Missing EF or NT-proBNP defaults to a documented healthy value (EF 62%, NT-proBNP 100 pg/mL), always flagged and never silently blended with measured values. A continuous-state mode serializes the Pulse engine state after each day and resumes from it the next day, instead of rebuilding the patient from scratch (Section VI).

TABLE VII
ENGINE AND MODELING PROBLEMS, ROOT CAUSES AND SOLUTIONS

Problem	Root cause (how found)	Solution	Evidence
Pulse rejected scenario files	Actions/conditions need PatientAction/AnyCondition wrappers; found from protobuf parse errors	Builders emit the wrapped schema	All 5 scenarios run
Runs exited 1 with time mismatch	CardiovascularMechanicsModification without Incremental:true silently re-stabilizes, consuming extra time	Set Incremental:true	Clean exit, exact end time
Collapse at low EF (negative chamber volume)	Binary EF condition stacked with a second EF-derived cut	Continuous multiplier carries only the additional acute severity	Collapse no longer occurs
Real patients outside Pulse range	Engine hard-rejects age > 65, BMI > 30; real HF cohorts are older	Clamp demographics only at the engine boundary; flag per patient	16/16 PerHeart patients flagged
Deconditioning looked like cardiac stress	Both used Exercise; HR 150–164 in both	Removed exertion from deconditioning	Distinct HR/MAP profiles (Table XII)
Fluid overload always scored LOW	Risk only measured change within a run; fluid overload is a shifted baseline	Added baseline-deficit term with max()	30/30 LOW → 29/30 MODERATE; scenario later found mis-specified
Live severity error 0.271 vs. 0.048 offline	NYHA ordinal used in training but unknown at inference, defaulted to class I	Removed NYHA from features; retrained	Live MAE 0.271 → 0.026
Engine crashes at high severity	Disease modifiers + sustained exertion exceed engine's sustainable load (Table VIII)	Characterized; failures recorded, never dropped	Deterministic at 154.1–154.3 s
Resumed state drifted	Disease modifiers are not carried by serialized state	Re-issue modifiers on every resume	Drift 6.5% → 0.000%
Engine segfault in continuous mode	Pulse Python SDK initialize_engine() crash (gdb trace); earlier “thread conflict” diagnosis disproved	Switched to the CLI scenario driver	Day 1→3 runs clean
Resumed runs flagged as crashes	Driver's end-time check ignores elapsed state time; schema field renamed (Mode, not Type)	Resume-aware crash filter; corrected schema	Exact match vs. continuous
SpO ₂ output always 0.0	Persists with Pulse's bundled patient and a request byte-identical to a known-good one; likely arm64 emulation	Never used as a feature	Excluded
ECG looked computed	Pulse ECG is a stored template scaled by HR	Labeled display-only in UI/API	Caveat in every response

VI. ENGINEERING CHALLENGES AND SOLUTIONS

Using Pulse as an HF twin surfaced problems that were not visible from documentation or offline evaluation. Table VII lists those specific to the engine and the modeling; general software and infrastructure issues are in Appendix Table XX.

A. Pulse instability at high severity

In a 150-patient batch (30 per scenario), 117 runs completed (Fig. 2). All 33 failures occurred in the two scenarios that add an *Exercise* action; 30 were engine crashes and 3 were timeouts, and failed patients had high severity (mean 0.75 and 0.72), so missingness is *not at random* with respect to severity. Engine logs for three independent crashes showed an identical chain: fatigue, hypoxia and tachycardia, then renal hypoperfusion and cardiovascular collapse, ending in a fatal negative heart-chamber volume and an irreversible-state event at 154.1–154.3 s. Table VIII lists single-variable interventions on a reproducible case. Gaps and ramps only delayed the crash; either stressor alone completed. The failure is therefore a sustained combined-load boundary whose safe exercise intensity varies by patient (about 0.22 for one patient and between 0.125 and 0.25 for another), so no single safe constant exists.

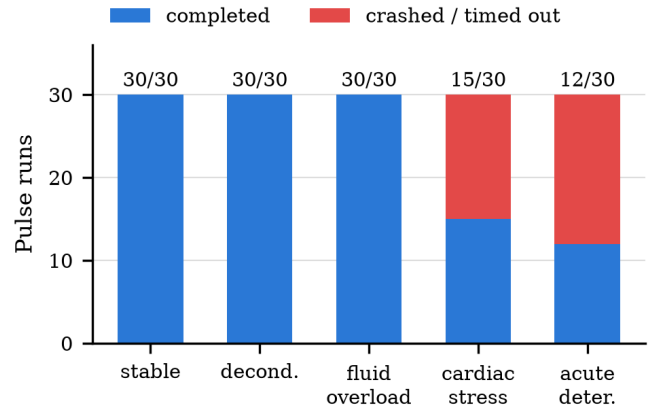


Fig. 2. Pulse batch completion by scenario (150 runs). All failures occur in the two scenarios that combine disease modifiers with an *Exercise* action.

B. Two failure mechanisms with opposite character

Failures were not a single phenomenon. Mechanism 1 (engine crash) depends on the patient's own severity and is deterministic, so retrying an identical scenario does not help. Mechanism 2 (timeouts and connection-pool exhaustion) depended on how many other patients were running; at 9

TABLE VIII
ISOLATION EXPERIMENTS ON A REPRODUCIBLE CRASH (SEVERITY 0.731)

Intervention	Outcome
Control (exact reproduction)	Crash at 154.32 s
60 s gap before exertion	Crash at 212.02 s
120 s gap before exertion	Crash at 272.92 s
Exertion ramped in 4 steps	Crash at 301.14 s
Exertion alone, no disease modifiers	Completes (660 s)
Disease modifiers alone, no exertion	Completes (660 s)
Exertion intensity halved	Completes (660 s)

TABLE IX
STATE RESUME FIDELITY (PATIENT EF 56.4, SEVERITY 0.46; FINAL VALUES AT 240 S)

Run	HR	MAP	CO	SV
Continuous (reference)	74.13	95.22	6.145	82.89
Resumed, modifier not re-issued	71.51	95.34	5.743	80.32
difference	-3.5%	+0.1%	-6.5%	-3.1%
Resumed, modifier re-issued	74.13	95.22	6.145	82.89
difference	0.000%	0.000%	0.000%	0.000%
Two continuous runs (control)	identical			

workers, 5/11 patients hit connection timeouts and 6/11 hit the 180 s engine timeout; at 4 workers 8/11 still timed out; at 2 workers there were no failures. A bare completion rate conflates the two; we report the mechanism with each rate.

C. Continuous state: carrying a patient forward

Rebuilding a patient every day discards the physiological state the twin is meant to track. Pulse can serialize its full state (about 2.33 MB, saved in 0.31–0.44 s and loaded in 0.43 s) and resume it, and with an `Exercise` action resumption was bit-for-bit identical to an uninterrupted run. With the disease modifier used by our pipeline it was not (Table IX): the modifier is not carried by the serialized state. Re-issuing it on every resume restored an exact match, which also requires storing the last-used EF and severity alongside the state. A later end-to-end test crashed inside the Pulse Python SDK; isolating it showed a native crash in the SDK’s engine initialization even with no machine-learning code loaded, disproving an earlier thread-conflict diagnosis, so the feature was rebuilt on the command-line driver.

D. Other modeling corrections

Three further corrections followed from running the real system rather than offline checks. (i) Severity (classifier output) and risk score (simulation-derived) share a $[0, 1]$ range but not a scale: for acute deterioration the risk score’s minimum (0.491) exceeded the severity mean (0.385). The projection module had applied a risk-score calibration to severity; the two calibrations were separated and the two quantities are now reported side by side, never fused. (ii) The classifier’s label occasionally flipped between days for genuinely ambiguous trajectories; a categorical persistence rule (window chosen by sweep) now stabilizes the reported scenario. (iii) An attempt to add rolling-window curvature features (“Option A”) kept

accuracy flat (90.7%) but worsened severity MAE from 0.0473 to 0.0506 on held-out data, so it was reverted.

VII. EVALUATION

We evaluate in layers of increasing realism. Table X summarizes what each layer tests and what it does not.

A. Offline model accuracy

On the 300-patient held-out test set, the classifier reached 90.7% accuracy (macro-F1 0.91) and the regressor severity MAE 0.047 (RMSE 0.061). Because test patients come from the same generator, these figures measure how well the model inverts the generator, not how it performs on independent trajectories; the pre-specified cohort below (17.6% day-by-day label accuracy) is the more relevant estimate. Cardiac stress (preserved EF) and acute deterioration (reduced EF), the pair hardest to separate from simulated HR/MAP alone, were cleanly separated because EF is an input. Removing all 24 wearable features cost 40 points of accuracy (Table XI), confirming that the wearable trend, not the clinical snapshot, carries most of the signal. The earlier 92.3% figure belonged to a version that included NYHA class, removed for the reason in Table VII.

B. Engine-level scenario validation

Each scenario was simulated for one patient at severity ≈ 0.5 (Table XII). Responses were physiologically distinct. The clearest contrast is between cardiac stress and acute deterioration: both raised heart rate, but the healthy heart (EF 67%) raised cardiac output from 5,910 to 9,948 mL/min with stroke volume maintained, while the failing heart (EF 26.6%) could not increase stroke volume (63→67 mL) and MAP fell to 52 mmHg. This contrast was not tuned; it emerged from the EF-driven modifiers. The engine’s left-heart pressure–volume signal was also physiologically plausible (volume 62.2–143.0 mL, pressure 5.8–136.8 mmHg at EF 45%) and is shown on the dashboard as a PV loop. Fig. 3 shows loops recorded during the scenario test for the same simulated patient before and after deterioration: by day 14 the loop has shifted to larger volumes (end-diastolic volume from about 150 to 220 mL) with a lower peak pressure (about 139 to 121 mmHg). The ECG beside it is a stored template that Pulse scales with heart rate; it does not respond to EF or scenario and is displayed for reference only.

C. Live API validation

Twenty-five synthetic patients (five per scenario) were re-played day by day through the live API with no mocking: 25/25 completed and all 25 were classified correctly, but severity MAE was 0.271, far from the offline 0.048. Per-patient inspection showed predictions compressed into a narrow 0.02–0.19 band, traced to the NYHA train/inference skew. After removing the feature and retraining, live re-validation reached severity MAE 0.008 on five patients and, expanded to 50 attempted (45 completed), MAE 0.0264 (95% bootstrap CI 0.0194–0.0352) with 100% scenario accuracy. The five

TABLE X
EVALUATION LAYERS AND THEIR SCOPE

Layer	Data	Main result	Does not test
Offline model test	300 held-out synthetic patients	90.7% accuracy, severity MAE 0.047	Real patients
Engine scenario check	5 synthetic patients in Pulse	Physiologically distinct scenario responses	Clinical accuracy
Live API validation	25 then 45 synthetic patients via live API	25/25 completed; live MAE 0.026 after fix	Real patients
Pre-specified scenario test	10 scripted stories $\times$ 6 seeds $\times$ 21 days	3/5 stories detected in all seeds; label accuracy 17.6%	Real deterioration events
Ablation (same runs)	30 deteriorating, 18 stable series	Tuned classifier matches or beats the Pulse-based alert	Clinician interpretability
Real-patient replay	PerHeart, 16 HF patients	13/16 completed end to end	Outcomes; patients' own physiology (age proxy)
BCG hemodynamics	2 patients, measured SV/CO	Simulated CO 45–46% below measured	Cohort-level accuracy; trends
Real-outcome tests	MIMIC-IV, Zigong	Twin's own scores near chance	Day-by-day early warning

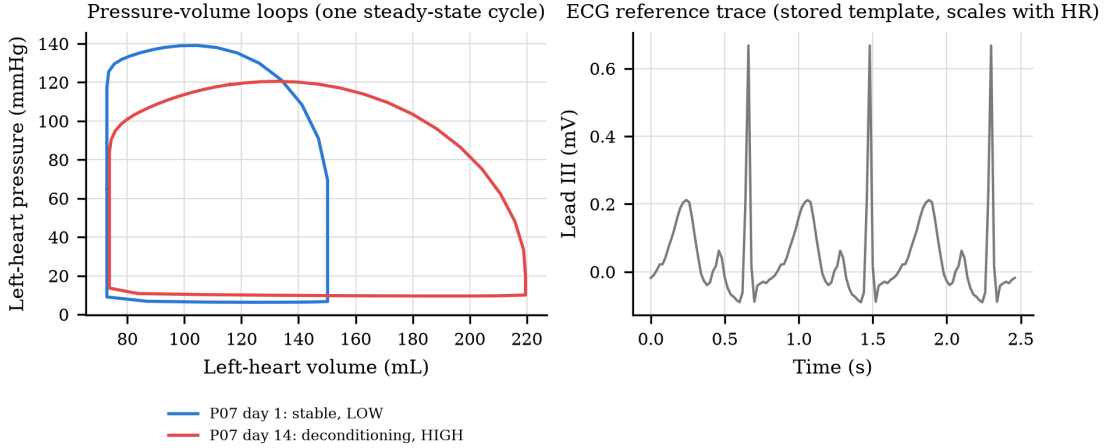


Fig. 3. Pulse output saved during the scenario test (P07, seed 45). Left: left-heart pressure–volume loop over one steady-state cardiac cycle on day 1 (LOW risk) and day 14 (HIGH risk). Right: the engine’s ECG reference trace, a stored waveform scaled by heart rate rather than a computed electrophysiological signal.

TABLE XI
SCENARIO MODEL ON 300 HELD-OUT SYNTHETIC PATIENTS

Variant	Accuracy	Severity MAE	RMSE
With NYHA feature (superseded)	92.3%	0.048	0.063
Production (clinical + wearable)	90.7%	0.047	0.061
+ curvature features (reverted)	90.7%	0.051	0.065
Clinical features only	50.3%	0.171	0.219

TABLE XII
SIMULATED RESPONSE BY SCENARIO (SEVERITY $\approx$ 0.5)

Scenario	HR (bpm)	MAP (mmHg)	Note
stable	71→70	95→95	Flat
deconditioning	71→74	95→95	Mild drift
fluid_overload	72→72	77→77	Congested baseline; CO +13%
cardiac_stress	71→164	95→67	Healthy compensation (EF 67%)
acute_deter.	72→132	78→52	SV cannot rise (EF 26.6%)

failures were all high-severity exertion scenarios, matching

Mechanism 1.

D. Pre-specified 21-day scenario test

Ten patient stories were written and their expected outcomes committed to the project’s git repository before any simulation (commit 38c3520, 3 October 2026, 02:04 IST; the first pilot run followed about 50 minutes later). This is pre-specification under version control, not registration in a public registry. The cohort (Table XIII) has five stories that should be caught, three that should stay quiet and two edge cases. Each story was run for 21 days in continuous-state mode on development seeds 42–44 and, after a rule was selected, on held-out seeds 45–47. A pilot run exposed a design confound (EF matched to story severity) and two harness bugs (day-to-day noise scaled to the current value instead of the population SD, and per-process salting of Python’s `hash()` making seeds non-reproducible); all were fixed and the harness was frozen before the main runs, with every amendment recorded.

Two alert definitions. The first analysis alerted on the classifier’s severity (> 0.15) and caught all five deteriorating patients, but also alerted on all three stable ones (58.7 false alerts per 100 patient-days). A code audit then showed that

TABLE XIII
PRE-SPECIFIED SCENARIO COHORT

ID	Story	Group
P01	Stable patient, ordinary life	stay quiet
P02	Salty weekend (weight bump, recovers)	edge case
P03	Slow, quiet weight gain	edge case
P04	Fluid overload building (sev. 0.20→0.50)	catch
P05	Gradual deconditioning (sev. 0.20→0.35)	catch
P06	Cardiac stress, exertion episodes	catch
P07	Sudden deterioration, days 10–13	catch
P08	Stressful fortnight, healthy heart	stay quiet
P09	Active, stable patient	stay quiet
P10	Everything goes wrong (sev. 0.20→0.45)	catch

the dashboard did not use this signal at all: it displayed the risk-score bucket (HIGH). Re-analysed under the dashboard’s definition, the system detected three of five deteriorating patients (P06, P07, P10) with 18.0 false alerts per 100 patient-days. P05 (deconditioning) produced near-zero values on every risk component on all 63 patient-days, and P04 (fluid overload) did not congest in some seeds, so neither had any signal for an alert rule to act on.

Alert rule selection. Three candidate rules were specified before computing any of them (Table XIV). C3 downgrades a sustained HIGH that is driven only by the baseline-deficit term with no corroborating instability, the diagnosed cause of P08’s persistent false alarm. It kept all three detections and cut false alerts by 67.6% on development seeds and 61.4% on held-out seeds, where it was never tuned.

Unified alert decision. The two alert signals were then merged into a single `decide_alert()` function, which is now the only source of alert state in the API and dashboard. It has three levels: ALERT (risk bucket HIGH and not downgraded by C3, or a crashed/failed engine run via the classifier-severity fallback); WATCH (risk bucket MODERATE, or a HIGH streak that C3 downgrades); and NONE. C3 therefore downgrades but never silences: a HIGH risk bucket can never map to NONE. Replaying the saved scenario-test runs through the deployed functions reproduced the detection pattern on both seed sets (P06, P07 and P10 at ALERT in every seed; P01 and P09 quiet in every seed). P08’s false alarm, previously ALERT for 8–14 consecutive days, was reduced to about 3 ALERT days per seed, with the remaining elevated days shown as WATCH. For P04 held-out seed 47, which never reached HIGH, 20 of 21 days now read WATCH, making a previously invisible fluid-overload trajectory visible for the first time.

Ablation: does the physiology layer improve alerting? The premise of the twin is that simulating physiology adds something the classifier alone does not. We tested this on the same saved runs by changing only the alert definition (Table XV, Fig. 4). Series-level sensitivity counts deteriorating patient-seed series with at least one alert on or after the story’s onset day; specificity counts stable series with no alert at all. At its default threshold (severity above the stable range, 0.15) the classifier detected all 30 deteriorating series but alerted in every stable series, and passing its output through Pulse

TABLE XIV
ALERT RULES: DETECTIONS AND FALSE ALERTS

Rule	Dev. (42–44)		Held-out (45–47)	
	Caught	FA/100	Caught	FA/100
Baseline (risk bucket HIGH)	3/5	17.99	3/5	30.16
C1: HIGH for 3 consecutive days	3/5	13.76	3/5	25.40
C2: add sustained-MODERATE watch	3/5	17.99	3/5	30.16
C3: component guard	3/5	5.82	3/5	11.64

TABLE XV
ALERT RULES ON THE SAME RUNS (ALL SIX SEEDS)

Alert rule	Sensitivity	FA / 100 pd
Weight rule	2/30 (0.02–0.21)	0.0
Classifier, severity > 0.15	30/30 (0.89–1.00)	60.1 (44.4–73.8)
Classifier, severity > 0.55 ^a	30/30 (0.89–1.00)	8.5 (4.0–13.8)
Classifier, severity > 0.60 ^a	28/30 (0.78–0.98)	2.4 (0.0–5.6)
Label only (exertion scenario)	19/30 (0.46–0.78)	18.8 (7.7–31.0)
Pulse risk bucket HIGH	21/30 (0.52–0.83)	24.1 (10.6–38.6)
Deployed (C3 guard)	21/30 (0.52–0.83)	8.7 (3.4–16.1)
Deployed, WATCH or ALERT	22/30 (0.56–0.86)	29.1 (14.6–46.3)

Wilson 95% CIs for sensitivity; bootstrap 95% CIs (2,000 resamples over series) for false alerts per 100 patient-days in the stable stories.
^aThreshold chosen with knowledge of these data; see Table XVI for a development/hold-out comparison.

and the C3 guard reduced false alerts from 60.1 to 8.7 per 100 patient-days at a sensitivity of 70%. That comparison, however, favors the physiology layer, because the classifier’s threshold was never tuned. Sweeping the classifier’s severity threshold traces a curve that lies above and to the left of every Pulse-based rule (Fig. 4): at 0.55 it detected 30/30 series with 8.5 false alerts per 100 patient-days, and at 0.60 it detected 28/30 with 2.4. A rule using only the predicted label (alert on an exertion scenario) performed similarly to the Pulse risk bucket (19/30 series, 18.8 false alerts), consistent with the label, more than the simulated physiology, determining when Pulse reports high risk.

To avoid tuning on the test data, we selected the classifier threshold on development seeds alone, using a rule fixed in advance (the lowest threshold whose development false-alert rate does not exceed the deployed system’s, 5.8), exactly as C3 was selected, and evaluated it once on held-out seeds (Table XVI). The rule chose 0.55. On held-out seeds the classifier detected all 15 deteriorating series and the deployed Pulse system 10, at the same false-alert rate (11.6 per 100 patient-days). In this case study, therefore, the physiology layer did not improve alerting over a properly thresholded classifier. A conventional weight rule (≥ 2 kg within 3 days) never false-alerted but detected only 2/30 series, because the scripted fluid build-up (about 0.4 kg per day) never reaches 2 kg within three days.

Absolute burden and sample size. Even the best rules carry a clinically heavy absolute rate: 8.7 false alerts per 100 patient-days means roughly one false alert every 11 or 12 days for each stable patient, and one every 9 days on

TABLE XVI
CLASSIFIER THRESHOLD SELECTED ON DEVELOPMENT SEEDS, TESTED ON HELD-OUT SEEDS

Held-out seeds 45–47	Sensitivity	FA / 100 pd
Classifier, severity > 0.55 (dev-selected)	15/15 (0.80–1.00)	11.6 (4.2–20.1)
Deployed Pulse system (C3, dev-selected)	10/15 (0.42–0.85)	11.6 (3.2–25.4)

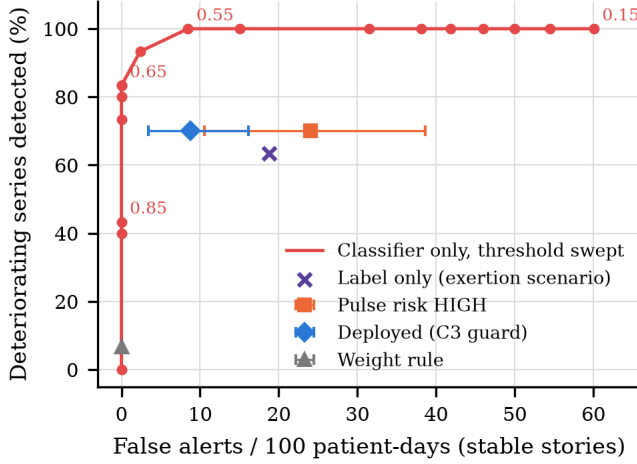


Fig. 4. Sensitivity versus false-alert burden on the same saved runs. The red curve sweeps the classifier’s severity threshold (labels mark thresholds); points show the Pulse-based rules (95% CIs), a label-only rule and a weight rule. The tuned classifier dominates the Pulse-based rules on this cohort.

the held-out seeds alone. With ten stories and six seeds, the confidence intervals are wide (Table XV); these results should be read as a controlled case study, not as estimates of performance in a patient population. Engine failures did not distort them: only 3 of 1,260 patient-days failed, and every metric was identical whether failed days were scored as alerts (the deployed fallback) or as no alert.

Why false alerts occur. Feature importances show the severity regressor relies on resting-HR delta and slope (63.2% combined) and step-count trends (27.2%), the signals that ordinary day-to-day variation moves most, while EF ranks tenth (0.6%). Adding Gaussian noise at one population SD to a stable patient’s window moved predicted severity from 0.11 to 0.43, crossing the stable-range threshold (Table XVII). **The classifier mostly simulated the wrong scenario.** Day-by-day scenario-label accuracy on this cohort was 17.6% (222/1,260 patient-days; 19.8% if days before a story’s onset are scored as “stable”), below the 20% expected from guessing among five classes and far below the 90.7% in-distribution test accuracy. The confusion matrix (Fig. 5) shows why: “deconditioning” was predicted on 71.5% of all days, including 480 of 630 days of stable stories and 113 of 126 days of the fluid-overload story; acute deterioration was recognized on only 11 days. A likely cause is distribution shift: the cohort’s neutral EF band (48–57%) sits below the EF range of the training

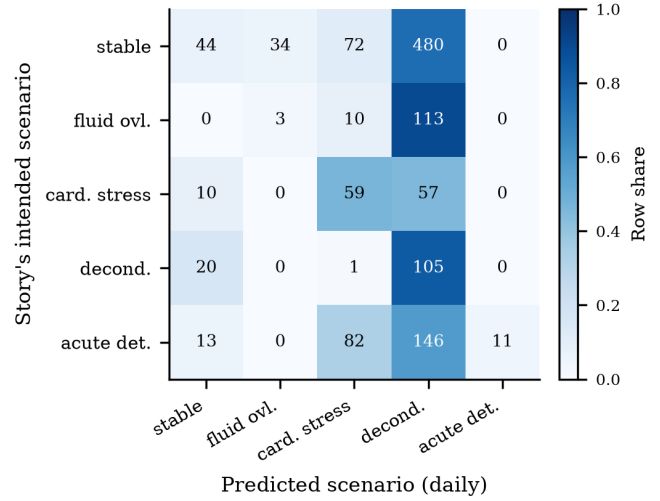


Fig. 5. Daily scenario predictions against each story’s intended scenario, all 1,260 patient-days (counts; colour is row share). Deconditioning dominates every row.

TABLE XVII
SENSITIVITY OF PREDICTED SEVERITY TO ADDED WEARABLE NOISE

Patient	0×SD	1×SD	1.5×SD	2×SD
P01 (stable)	0.115	0.428	0.544	0.560
P10 (deteriorating)	0.777	0.570	0.622	0.850

“stable” class (51.3–70.7%) and near the deconditioning mean. Because the predicted label chooses which Pulse actions run, the engine simulated a scenario other than the intended one on most days. The risk score still reached HIGH on 28% of deconditioning-labeled days (255/898) and 83% of cardiac-stress-labeled days (185/224); inspection of the stored runs shows that once exertion days have occurred, the carried-forward engine state keeps the score high on later days even when the label reverts to deconditioning (Fig. 6). The ablation results below therefore describe the pipeline as it behaves with these label errors, not a physiology layer driven by correct scenarios.

E. Real-patient replay (PerHeart)

The PerHeart pilot dataset [25] contains one month of real home measurements (pulse oximeter, scale, blood-pressure cuff) from 27 HF patients aged 67–94. Sixteen had at least 21 real overlapping days of heart rate, SpO₂ and weight; no day was fabricated, and unmeasured fields (sleep, HRV, steps, EF, NT-proBNP) took documented defaults with explicit flags. Every patient required the engine’s age proxy. Before the severity fix, all 16 completed but severities were compressed (0.078–0.148) and no patient reached HIGH, independently reproducing the NYHA skew on real data. After the fix, severities spread (0.093–0.516) and risk buckets were 4 LOW, 4 MODERATE and 5 HIGH (previously 4 LOW, 12 MODERATE, 0 HIGH), while three patients now crashed with the deterministic engine signature. All 16 were staged NYHA I,

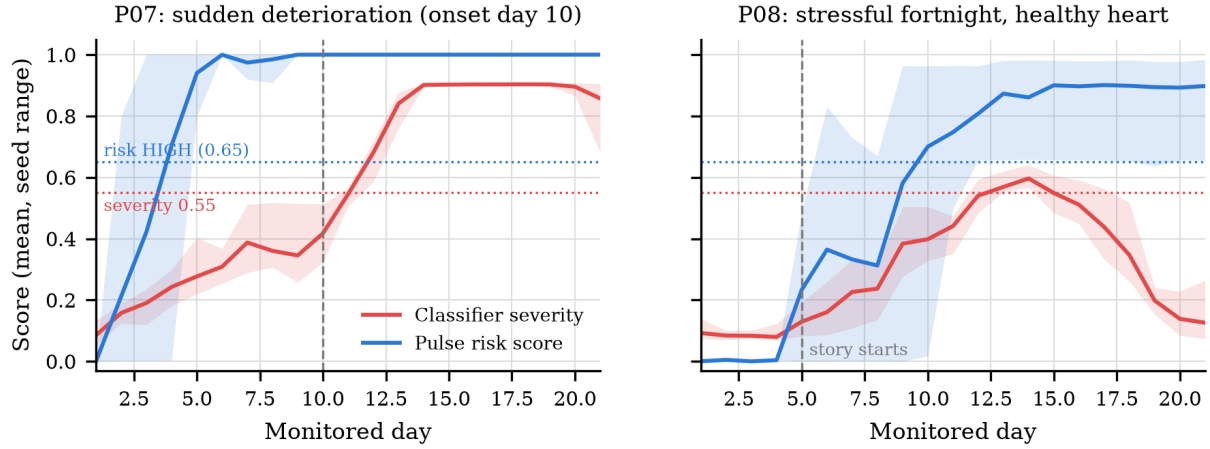


Fig. 6. Daily classifier severity and Pulse risk score (mean and range over six seeds) for a deteriorating story (P07) and a stable story with a stressful fortnight (P08). In P07 the risk score saturates days before the designed onset; in P08 it stays high after the stress episode ends while classifier severity recovers.

TABLE XVIII
SIMULATED VS. MEASURED HEMODYNAMICS (RATIO SIM/REAL)

Subject	HR	Stroke volume	Cardiac output
14	0.94	0.58	0.54
102	0.63	0.87	0.55

an artifact of the identical healthy EF/BNP defaults rather than a clinical finding. The one fluid-overload patient scored LOW because the EF default makes Pulse simulate a structurally normal heart, so the baseline-deficit term cannot fire; the API now names this mechanism in its caveat. We searched the literature for a BNP-to-EF proxy and found none: the clinical relationship runs from EF to BNP, not the reverse.

F. Measured hemodynamics (BCG, two patients)

For two patients from a ballistocardiography dataset with echocardiographic volumes [27], BCG features adjusted the twin’s vascular compliance and simulated stroke volume and cardiac output were compared with measured values (Table XVIII). Simulated cardiac output was 45–46% below measured in both; 71.3% of patient 14’s stroke-volume shortfall was already present from anthropometrics alone, before EF or BCG data entered the model. A synthetic 21-day deterioration anchored to these patients’ real HR and EF was eventually classified correctly for both, but patient 14 was labeled fluid overload for days 6–10 before converging.

G. Real-outcome tests

Two real cohorts lack the 21-day wearable window and EF needed to run the whole twin, so only components with a non-fabricated input path were tested (Fig. 7, Table XIX). The baseline-deficit term is a function of starting MAP alone; applied to measured first-24-hour MAP in 17,129 MIMIC-IV HF ICU admissions, it predicted in-hospital mortality with AUC 0.596 (95% CI 0.585–0.608), with mortality rising from

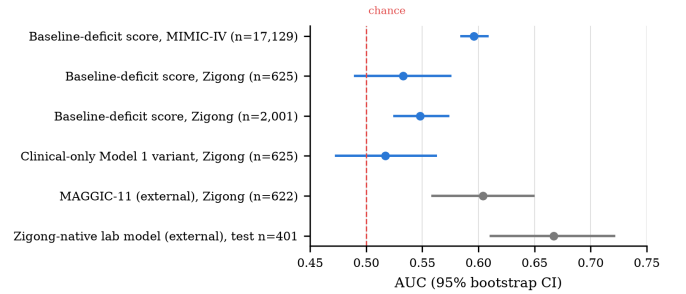


Fig. 7. Real-outcome discrimination. Blue: components of the twin. Grey: external benchmarks on the same cohort.

8.9% to 21.2% across score deciles. This shows that low admission MAP carries modest prognostic information in an ICU population; it is not a test of the twin, which was not run on these patients. On the Zigong cohort [26] (6-month death or readmission), the same term and a clinical-only variant of the scenario model were indistinguishable from chance. As external context, published LACE-index AUCs [18] lie around 0.56–0.65 in other populations and outcomes (not directly comparable), an 11-of-13-variable MAGGIC score reached 0.604 on this cohort, and a model using laboratory panels reached 0.667, signal carried by renal and coagulation tests that a wearable-based system cannot obtain.

On the synthetic batch, the risk score agreed moderately with MAGGIC computed on the same patients (Spearman $r = 0.479$; tier agreement 38.5% against $\approx 33\%$ by chance), indicating directional consistency rather than validity.

VIII. DISCUSSION

What the twin does well. The pipeline runs end to end, day after day, on synthetic and real home data; it separates physiologically different presentations through mechanism rather than labels (Table XII); it explains alerts in terms of simulated hemodynamics; and its alert rule generalized

TABLE XIX
REAL-OUTCOME TESTS (AUC, 95% BOOTSTRAP CI)

Score / cohort	<i>n</i>	AUC
Baseline deficit, MIMIC-IV, in-hospital death	17,129	0.596 (0.585–0.608)
Baseline deficit, Zigong, complete-case	625	0.533 (0.490–0.575)
Baseline deficit, Zigong, all cleaned	2,001	0.548 (0.525–0.573)
Clinical-only scenario model, Zigong	625	0.517 (0.473–0.562)
MAGGIC-11 (external) [17], Zigong	622	0.604 (0.559–0.649)
Zigong lab model (external), test set	401	0.667 (0.611–0.721)

Zigong outcome: death or readmission within 6 months (41.3% event rate).
Bootstrap: 2,000 resamples.

from development to held-out seeds. Continuous state makes the twin a persistent model of a patient rather than a fresh simulation each day.

Is the mechanistic layer circular? Largely, in its current form. The Pulse scenario is selected by the classifier’s label and scaled by its severity, so the simulated physiology re-expresses the classifier’s decision through a body model rather than observing new evidence. The ablation confirms this: a label-only rule behaves like the Pulse risk bucket, and a classifier threshold tuned on development seeds matches the deployed system’s held-out false-alert rate while detecting more deteriorating series. Two factors limit what the engine could add here. First, the label was wrong on most days, so the engine usually simulated the wrong scenario. Second, the scenario that should carry congestion (fluid overload) is mis-specified. What the engine does provide is an interpretable hemodynamic account of an alert (pressure–volume loops, cardiac output, compensation), whose value to clinicians has not yet been tested. Showing an alerting benefit would require scenario labels that are correct, a corrected fluid-overload scenario, and ideally physiological inputs (e.g., measured weight-driven volume changes) that do not pass through the classifier.

How personal is the twin? Less than the term usually implies. On real patients, every PerHeart body was an age proxy, ejection fraction and NT-proBNP were defaults, simulated cardiac output was 45–46% below measured in the two BCG patients, and the severity regressor relies on resting-HR and step trends for about 90% of its importance (EF: 0.6%). The current system is best described as a population-parameterized physiological model driven by an individual’s wearable trends; per-patient calibration of baseline hemodynamics is the main step toward a personal twin.

What it does not yet do. Five findings bound the claims. First, two of five deterioration types are invisible to the current alert because their signatures never reach the simulated hemodynamics: deconditioning by design (no acute event), and fluid overload because the scenario reduces venous compliance without adding circulating volume, which mobilizes pooled blood rather than overloading the circulation. Second, the models are sensitive to ordinary wearable noise and to clinical inputs outside their training distribution. Third, the twin’s own scores do not predict real outcomes beyond chance. Fourth,

the physiology layer did not improve alerting over a tuned classifier. Fifth, even the best configurations produce roughly one false alert every 9–12 days per stable patient, a burden that would be hard to sustain clinically. Our results therefore support the feasibility of running such a twin end to end, not an alerting benefit or clinical effectiveness.

Lessons for building physiology-engine twins. Every major fault in Table VII was found by running the real system, not by offline evaluation: the NYHA skew only appeared in live inference, the CORS fault only in a browser, the resume drift only with the production modifier, and the alert mismatch only by auditing what the dashboard actually displayed. Pre-registering expectations and holding out seeds kept fixes honest, and reporting failure mechanisms rather than bare completion rates prevented a biased reading of the results.

IX. LIMITATIONS

Training data are synthetic, although every parameter is traced to a real aggregate statistic, and the pre-specified cohort is small (ten stories, six seeds). The fluid-overload scenario contains a modeling error rather than a mere limitation: it lowers venous compliance without adding circulating volume, which mobilizes pooled blood and moves the simulated hemodynamics in the improving direction as severity rises. Correcting it requires a volume-loading action, re-verification of engine stability, re-running the scenario validation, retraining the severity model and repeating the 21-day test; we have not yet done so. Several scoring elements were tuned post hoc (Table VI). Pulse’s native range (age 18–65, BMI 16–30) excludes most HF patients, so older patients are simulated with a proxy body. Real EF and NT-proBNP were unavailable in the replay cohort, and their defaults mask the fluid-overload mechanism. The engine’s SpO₂ output was unusable and its ECG is a template. Pulse is unstable when disease modifiers combine with sustained exertion, which under-represents high-severity exertion scenarios in all datasets. Medication effects are not modeled: Pulse includes a validated diuretic (furosemide), making diuretic modeling feasible, but has no beta-blocker or ACE-inhibitor/ARB substances. Risk-bucket cut-points and the deterioration-rate calibration constants are engineering choices, not clinically validated thresholds. The API has no authentication and is suitable only for research use.

X. CONCLUSION AND FUTURE WORK

We built and evaluated a physiological digital twin for HF home monitoring that interprets daily trends through a mechanistic engine and carries simulated state from day to day. It runs end to end on synthetic and real home data and produces interpretable hemodynamics. In a pre-specified case study, however, its physiology layer did not improve alerting over the classifier with a properly selected threshold: on held-out seeds the classifier detected all deteriorating series at the same false-alert rate at which the Pulse system detected two thirds. The reasons are specific (label collapse to deconditioning, a mis-specified fluid-overload scenario, and dependence of the

TABLE XX
SOFTWARE AND INFRASTRUCTURE ISSUES

Problem	Root cause (how found)	Solution	Evidence
Timeouts at 180 s	Some runs need ≈ 180 s under arm64 emulation	Documented margin; 2-worker cap	Re-attempts at 180.3–180.4 s
Failures under concurrency	SQLAlchemy pool (5+10) exhausted; CPU contention	Concurrency $9 \rightarrow 4 \rightarrow 2$	0 failures at 2 workers
Corrupted Pulse image Python 3.9 container errors	Local containerd-snapshotter extraction bug PEP 604 unions evaluated at class definition by SQLAlchemy/Pydantic	Disabled snapshotter, re-pulled Optional[...] in model/schema files	Engine verified Imports succeed
Dashboard could not call API	Browser enforces CORS (invisible to curl/TestClient)	CORS middleware (local only)	Found in browser test
Model drift across machines	Unpinned scikit-learn/numpy versions	Exact dependency pins; model SHA-256 checked before each run	Zero version warnings

simulated scenario on the label) and therefore actionable. Future work will (0) fix the scenario labels, through training data closer to real trajectories or physiological inputs that bypass the label, and re-run the ablation; (1) add a volume-loading mechanism to the fluid-overload scenario and a diuretic action using Pulse’s furosemide model; (2) systematically map the engine’s stability boundary across severity, scenario and body profile; (3) make the models robust to wearable noise and out-of-distribution clinical inputs; (4) personalize the twin’s baseline heart rate and hemodynamics from measured data; and (5) evaluate day-by-day early warning on real longitudinal data with recorded HF events, the test that ultimately decides clinical value.

APPENDIX A

SOFTWARE AND INFRASTRUCTURE ISSUES

Table XX lists issues that were general to the software stack rather than to the engine or the model; they are reported for reproducibility.

DATA AND CODE AVAILABILITY

Synthetic data, derived results and code are in the project repository. Raw MIMIC-IV, Zigong and PerHeart files are not redistributed, in line with their access agreements; PerHeart-derived outputs are shared under CC BY-NC-SA.

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